1. Which of the following mathematical expression represents the rotated five-point formula for 2D Laplace partial differential equation for the uniform step size in both x and y direction?

A. $u_{i,j} = \frac{1}{4}[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$

B. $u_{i,j} = 4[u_{i-1,j-1} + u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j+1}]$

C. $u_{i,j} = \frac{1}{4}[u_{i-1,j-1} + u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j+1}]$

D. $u_{i,j} = 4[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$

2.

In Doolittle's method of LU factorization matrix, _____ has 1's on its diagonal.

A. U

B. LU

C. L

D. UL

3.

The sufficient condition for convergence of Jacobi's iterative method is that the system of linear equations must be ____.

A. linearly dependent

B. diagonal dominant

C. off-diagonal dominant

D. linearly independent

4.

Which of the following inequality is sufficient condition for the convergence of Jacobi's iterative method?

A. $|a_{ii}| \leq \sum_{i \neq j}^{n} |a_{i,j}| \quad , \ i = 1,2, \dots, n$

B. $|a_{ii}| \geq \sum_{i \neq j}^{n} |a_{i,j}| \quad , \ i = 1,2, \dots, n$

C. $|a_{ij}| \leq \sum_{i \neq j}^{n} |a_{i,i}| \quad , \ i = 1,2, \dots, n$

D. $|a_{ij}| \geq \sum_{i \neq j}^{n} |a_{i,i}| \quad , \ i = 1,2, \dots, n$

5.

Crank-Nicolson method when applied on parabolic PDE reduces the total _____.

A.

B. computational cost and computational time

C. computational cost

D. none of these

computational time

6.

Explicit Schmidt Scheme can be obtained for heat diffusion equation by applying forward difference on time-derivative and _____ on space-derivative.

A.

none of these

B.

central difference

C.

forward difference

D.

backward difference

---

7. Which of the following mathematical expression represents the standard five-point formula for 2D Laplace partial differential equation for the uniform step size in both x and y direction?

A.

$u_{i,j} = 4[u_{i-1,j-1} + u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j+1}]$

B.

$u_{i,j} = \frac{1}{4}[u_{i-1,j-1} + u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j+1}]$

C.

$u_{i,j} = \frac{1}{4}[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$

D.

$u_{i,j} = 4[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$

---

8. Fourier analysis method is used to find the ……. of finite difference scheme

A. Convergence

B. Consistency

C. Stability

D. None of these

---

9. Which of the following equation represents 2D Poisson partial differential equation?

A.

$$\nabla u = 0$$

B.

$$\nabla^2 u = 0$$

C.

$$\nabla^2 u = f(x, y)$$

D.

$$\nabla u = f(x)$$

10.

$$Consider\ the\ PDE\ \frac{\partial f(x,t)}{\partial t} + c\frac{\partial f(x,t)}{\partial x} = 0$$

$$which\ of\ the\ following\ difference\ approximation\ will\ be\ used\ to\ make\ upwind\ scheme\ when\ c > 0$$

A. Forward difference

B. Backward difference

C. Crank-Nicolson scheme

D. Central difference
